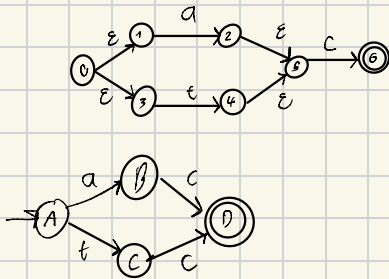


Ejercicio No. 1 (25%) – Convierta las siguientes expresiones regulares en autómatas finitos deterministas (para ello deberá primero convertir las expresiones regulares a AFN y luego convertir a AFD). Muestre todo su procedimiento, i.e., AFN construido con Thompson, tabla de transición, conversión a AFD. Para el inciso g, interprete \ como un escape de carácter, i.e., \ significa que su regex reconoce el carácter \.

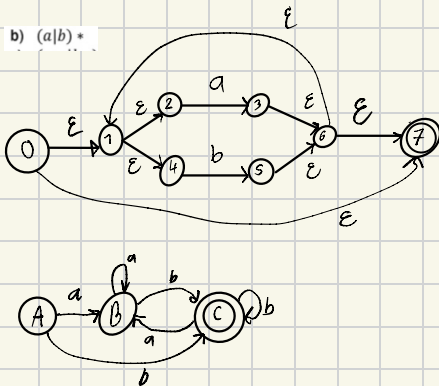
a) $(a|t)c$



	a	t	c	ε
0	✓	✓	✓	{0, 3}
1	2	✓	✓	{1}
2	✓	✓	✓	{2, 5}
3	✓	4	✓	{3}
4	✓	✓	✓	{4, 5}
5	✓	✓	6	{5}
6	✓	✓	✓	{6}

Estado AFN	aε	tε	cε*
A = {0, 3}	B	C	✓
B = {2, 5}	✓	✓	D
C = {4}	✓	✓	D
D = {6}	✓	✓	✓

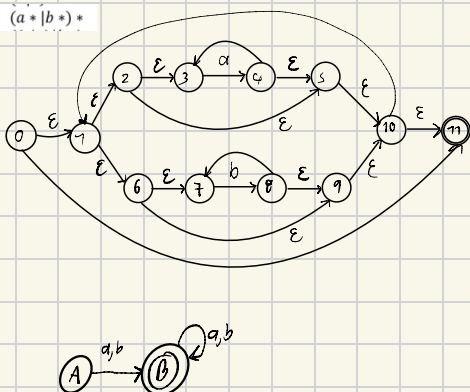
b) $(a|b)^*$



	a	b	ε
0	✓	✓	{0, 3}
1	✓	✓	{1, 3, 4}
2	3	✓	{2}
3	✓	✓	{3, 6}
4	✓	{5}	{4}
5	✓	✓	{3, 6}
6	✓	✓	{6, 7, 3}
7	✓	✓	{7}

Estado AFN	aε	bε*
A = {0, 3, 4, 7}	B	C
B = {1, 2, 3, 4, 6, 3}	B	C
C = {1, 2, 4, 5, 6, 3}	B	C

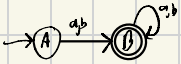
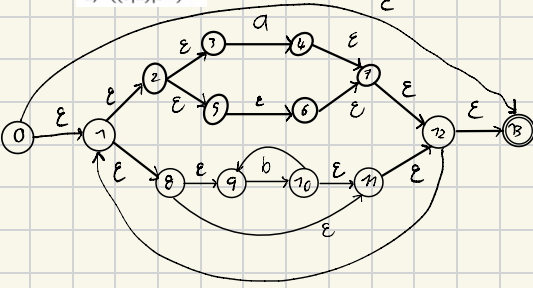
c) $(a^*|b^*)^*$



	a	b	ε
0	✓	✓	{1}
1	✓	✓	{1, 3, 7}
2	✓	✓	{2, 3, 5}
3	4	✓	{3}
4	✓	✓	{3, 4, 5}
5	✓	✓	{5, 10}
6	✓	✓	{6, 7, 9}
7	✓	8	{7}
8	✓	✓	{8, 7, 9}
9	✓	✓	{9, 10}
10	✓	✓	{10, 11, 3}
11	✓	✓	{11}

Estado AFN	aε	bε*
A = {0, 1, 3, 4, 5, 6, 7, 9, 10, 11}	B	B
B = {0, 3, 3, 5, 1, 3, 5, 7, 10, 11}	B	B

d) $((\epsilon|a)|b)^*$



Estado AFN

	$a\epsilon^*$	$b\epsilon^*$
$A = \{0, 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$	B	B
$B = \{0, 1, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$	B	B

	a	b	ϵ
0	/	/	$\{0, 1, 13\}$
1	/	/	$\{2, 3, 5\}$
2	/	/	$\{3, 5\}$
3	/	/	$\{3\}$
4	/	/	$\{4, 7\}$
5	/	/	$\{5, 6\}$
6	/	/	$\{6, 9\}$
7	/	/	$\{7, 12\}$
8	/	/	$\{8, 9, 11\}$
9	/	10	$\{9\}$
10	/	/	$\{9, 10, 11\}$
11	/	/	$\{11, 12\}$
12	/	/	$\{12, 13, 11\}$
13	/	/	$\{13\}$

e) $(a|b)^*abb(a|b)^*$

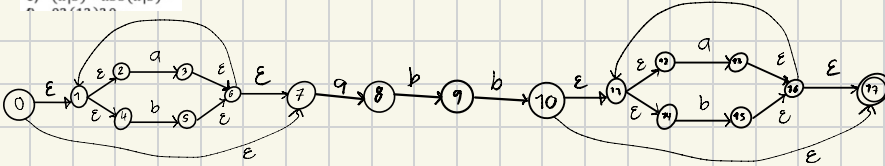
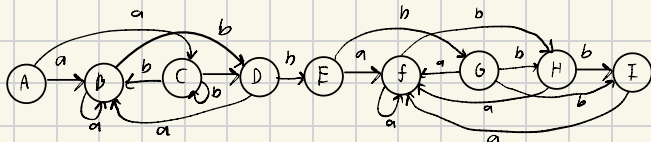


Tabla AFD

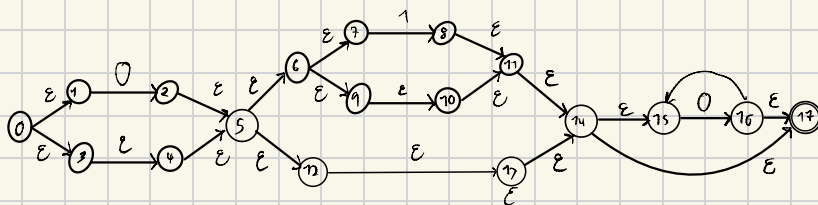
	a	b	ϵ
0	/	/	$\{0, 13\}$
1	/	/	$\{1, 3, 4\}$
2	3	/	$\{2\}$
3	/	/	$\{3, 6\}$
4	/	5	$\{4\}$
5	/	/	$\{5, 6\}$
6	/	/	$\{6, 7\}$
7	8	/	$\{7\}$
8	/	9	$\{8\}$
9	/	10	$\{9\}$
10	/	/	$\{10, 11\}$
11	/	/	$\{11, 12, 14\}$
12	13	/	$\{12\}$

Estados	conjuntos	$a\epsilon^*$	$b\epsilon^*$
A	$\{0, 1, 2, 4, 13\}$	B	C
B	$\{1, 2, 3, 4, 6, 7, 8\}$	B	D
C	$\{1, 2, 4, 5, 6, 13\}$	B	C
D	$\{1, 2, 4, 5, 6, 7, 9\}$	B	E
E	$\{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 15\}$	F	G
F	$\{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17\}$	F	H
G	$\{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$	F	H
H	$\{1, 2, 4, 5, 6, 7, 9, 11, 12, 14, 15, 16, 17\}$	F	I
I	$\{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 15, 16, 17\}$	F	G



7) $0?(1?)?0^*$

$(\epsilon|0)(\epsilon|(\epsilon|1))0^*$

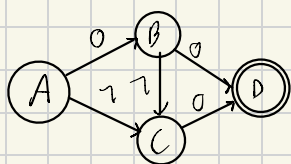


	0	1	ϵ
0	/	/	$\{0, 1, 3\}$
1	2	/	$\{1\}$
2	/	/	$\{2, 5\}$
3	/	/	$\{3, 4\}$
4	/	/	$\{4, 5\}$
5	/	/	$\{5, 6, 12\}$
6	/	/	$\{6, 7, 9\}$
7	/	8	$\{7\}$
8	/	/	$\{8, 13\}$
9	/	/	$\{9, 10\}$
10	/	/	$\{10, 11\}$
11	/	/	$\{11, 14\}$
12	/	/	$\{12, 17\}$

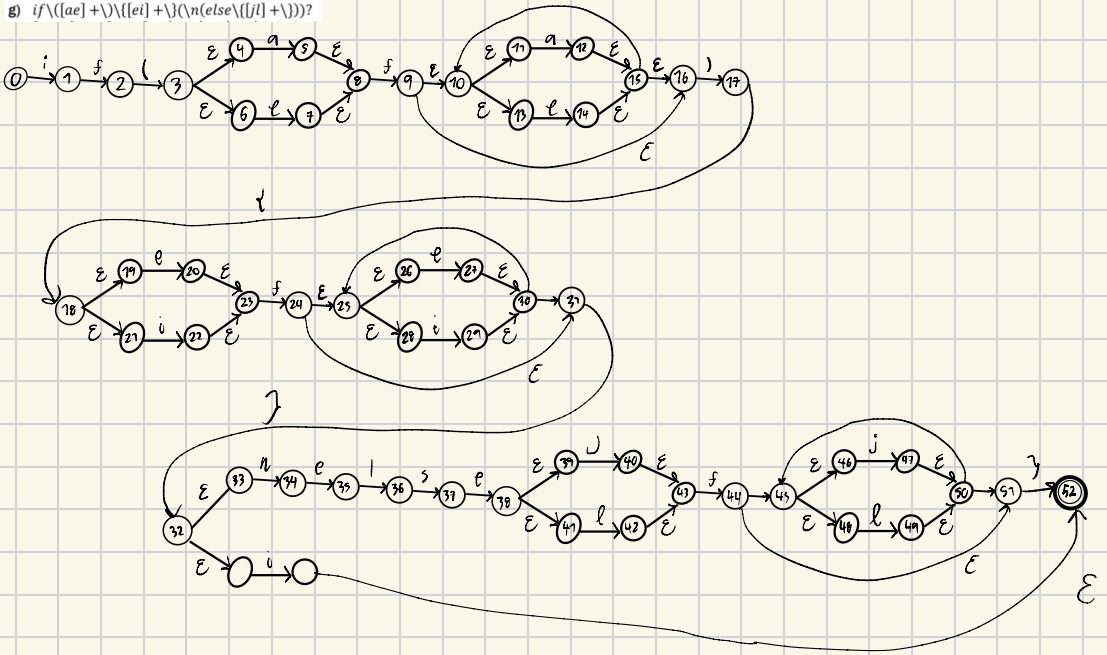
	0	1	ϵ
13	/	/	$\{13, 14\}$
14	/	/	$\{14, 15\}$
15	16	/	$\{15\}$
16	/	/	$\{16, 18\}$
17	/	/	$\{17\}$

Tabla AFD

Estados	Conjuntos	$0\epsilon^*$	$1\epsilon^*$
A	$\{0, 1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14\}$	B	C
B	$\{2, 5, 6, 7, 9, 10, 11, 12, 13, 14\}$	D	C
C	$\{8, 13, 14\}$	D	
D	$\{14\}$		

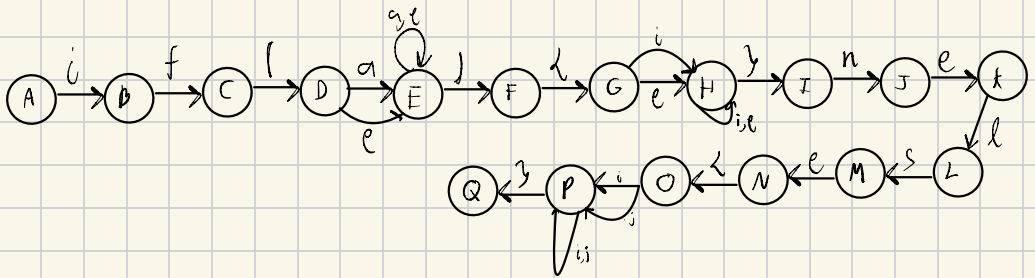


8) $i f \setminus ([ae] + \setminus) \setminus \{lei\} + \setminus) \setminus (n(else \setminus \{lil\} + \setminus))?$

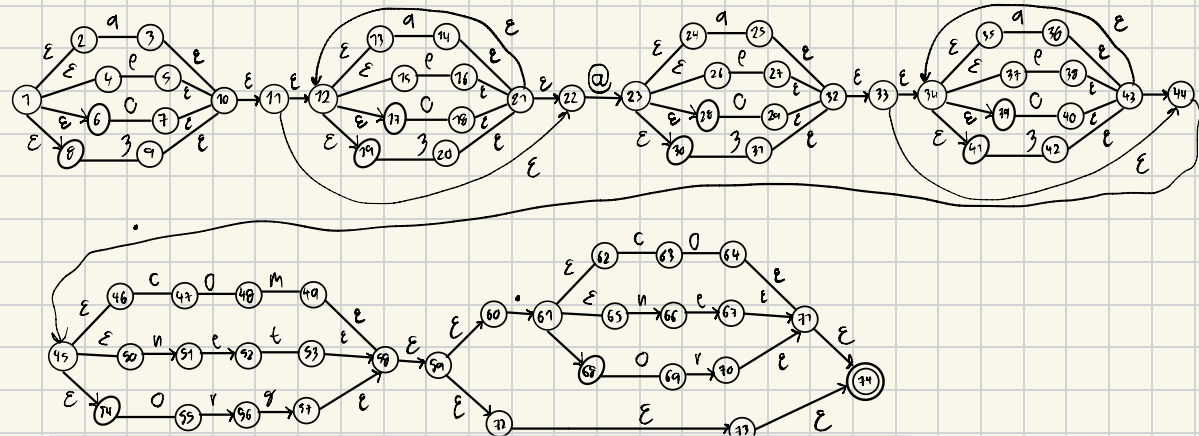


Estado	i	f	l	e	s	n	ε
0	1						0
1		2					1
2			3				2
3							3,4,6
4				5			4,8
5							5
6							6,7
7							7,8
8				9			8
9							9,10,16
10							10,11,13
11				12			11
12							12,15
13							13,14
14							14
15							10,15,16
16					17		16
17						18	17
18							16,19,21
19				20			19
20							20,23
21	22						21
22							22,23
23		24					23
24							24,25,31
25							25,26,28
26					27		26
27							27,30
28	29						28
29							29,30
30							25,30,31
31						32	31
32							32,33,52
33						34	33
34				35			34
35						36	35
36							36
37				38			37
38							38,39,41
39							39
40							40,43
41						42	41
42							42,43
43		44					43
44							44,45,51
45							45,46,48
46							46
47							47,50
48						49	48
49							49,50
50							45,50
51						52	51
52							52

Estado	Conjuntos	ie*	fe*	ce*	ae*	ee*	je*	ic*	jc*	nc*	lc*	se*	jc*	Final
A	[0]	B	C	C	C	C	C	C	C	C	C	C	A	No
B	[1]	C	D	C	C	C	C	C	C	C	C	C	B	No
C	[]	C	C	C	C	C	C	C	C	C	C	C	C	No
D	[2]	C	C	E	C	C	C	C	C	C	C	C	D	No
E	[3, 4, 6, 7, 8]	C	F	C	G	C	C	C	C	C	C	H	E	No
F	[9, 10, 11, 13, 14, 16]	C	C	C	I	J	C	C	C	C	C	K	F	No
G	[5]	C	C	C	C	C	C	C	C	C	C	C	G	No
H	[4]	C	C	C	G	C	C	C	C	C	C	H	L	No
I	[12, 15, 16]	C	C	C	C	J	C	C	C	C	C	C	M	No
J	[17]	C	C	C	C	C	N	C	C	C	C	C	J	No
K	[11]	C	C	C	I	C	C	C	C	C	C	K	C	No
L	[8]	C	F	C	C	C	C	C	C	C	C	C	L	No
M	[10, 11, 12, 13, 14, 15, 16]	C	C	C	I	J	C	C	C	C	C	K	M	No
N	[18, 19, 21]	O	C	C	C	P	C	C	C	C	C	Q	N	No
O	[22, 23]	C	R	C	C	C	C	C	C	C	C	C	O	No
P	[20, 23]	C	R	C	C	C	C	C	C	C	C	C	P	No
Q	[19]	C	C	C	C	P	C	C	C	C	C	Q	C	No
R	[24, 31]	C	C	C	C	C	C	C	C	S	T	R	U	No
S	[32, 33, 52]	C	C	C	C	C	C	C	C	C	C	C	S	Si
T	[31]	C	C	C	C	C	C	C	C	S	C	C	C	No
U	[25, 26, 28]	V	C	C	C	C	C	C	C	C	C	C	U	No
V	[29, 30, 31]	C	C	C	C	C	C	C	C	S	C	C	W	No
W	[25, 26, 28, 29, 30, 31]	V	C	C	C	C	C	C	C	S	C	C	W	No



h) `[ae03] + @[ae03]+.(com|net|org)(.(gt|cr|co))?`



Estados	a	c	0	3	@	c	e	m	a	t	g	c
1												1,2,4,6,8
2	3											2
3												3,10
4		5										4
5			7									5,10
6												6
7												7,10
8				9								8
9												9,10
10												10,11
11												11,12,22
12												12,13,15,17,
13	14											13
14												14,21
15		16										15
16												16,21
17			18									17
18												18,21
19				20								19
20												20,21
21					x							12,21,22
22					23							22
23												23,24,26,28,30
24	25											24
25												25,32
26			27									26
27												27,32
28				29								28
29												29,32
30												30
31				31								31,32
32												32,33
33												33,34,44
34												34,35,37,39,41
35	36											35
36												36,43
37		38										37
38												38,43
39			40									39
40												40,43
41												41
42				42								42,43
43												34,43,44
44												44,45
45												45,46,50,54
46							47					46
47								48				47
48									49			48
49												49,58
50									51			50
51			52									51
52										53		52
53												53,58
54								55				54
55										56		55
56											57	56
57												57,58
58												58,59
59												59,60,72
60						61						60
61												61,62,65,68
62											63	62
63										64		63
64												64,71
65							66					65
66										67		66
67												64,71
68						69						68
69								70				69
70												70,71
71												71,74
72												72,73
73												73,74
74												74

DFA State	NFA States	c	c	0	3	@	c	c	a	m	n	t	r	s	Final
D0	{1, 2, 4, 6, 8}	01	02	03	04	05	05	05	05	05	05	05	05	05	No
D1	{3, 10, 11, 12, 13, 15, 17, 19, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D10	{23, 24, 26, 28, 30}	011	012	013	014	05	05	05	05	05	05	05	05	05	No
D11	{25, 32, 33, 34, 35, 37, 39, 41, 44, 45, 4015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D12	{27, 32, 33, 34, 35, 37, 39, 41, 44, 45, 4015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D13	{29, 32, 33, 34, 35, 37, 39, 41, 44, 45, 4015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D14	{31, 32, 33, 34, 35, 37, 39, 41, 44, 45, 4015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D15	{33, 35, 37, 39, 41, 43, 44, 45, 46, 5015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D16	{34, 35, 37, 39, 41, 43, 44, 45, 46, 5015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D17	{34, 35, 37, 39, 40, 41, 43, 44, 45, 46, 5015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D18	{34, 35, 37, 39, 41, 42, 43, 44, 45, 46, 5015}	016	017	018	05	05	019	020	05	021	05	05	05	05	No
D19	{47}	05	05	05	05	05	05	022	05	05	05	05	05	05	No
D2	{5, 10, 11, 12, 13, 15, 17, 19, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D20	{55}	05	05	05	05	05	05	05	05	05	05	023	05	05	No
D21	{51}	05	024	05	05	05	05	05	05	05	05	05	05	05	No
D22	{48}	05	05	05	05	05	05	05	025	05	05	05	05	05	No
D23	{56}	05	05	05	05	05	05	05	05	05	05	05	026	05	No
D24	{52}	05	05	05	05	05	05	05	05	05	05	027	05	05	No
D25	{48, 58, 59, 60, 72, 73, 74}	05	05	05	05	028	05	05	05	05	05	05	05	05	Yes
D26	{57, 58, 59, 60, 72, 73, 74}	05	05	05	05	028	05	05	05	05	05	05	05	05	Yes
D27	{53, 58, 59, 60, 72, 73, 74}	05	05	05	05	028	05	05	05	05	05	05	05	05	Yes
D28	{61, 62, 65, 68}	05	05	05	05	05	029	05	05	05	05	05	05	030	No
D29	{66, 69}	05	05	05	05	05	05	031	05	05	05	05	032	05	No
D3	{7, 10, 11, 12, 13, 15, 17, 19, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D30	{63}	05	05	05	05	05	05	05	05	05	033	05	05	05	No
D31	{70, 71, 74}	05	05	05	05	05	05	05	05	05	05	05	05	05	Yes
D32	{64, 67, 71, 74}	05	05	05	05	05	05	05	05	05	05	05	05	05	Yes
D33	{64, 71, 74}	05	05	05	05	05	05	05	05	05	05	05	05	05	Yes
D4	{9, 10, 11, 12, 13, 15, 17, 19, 22}	05	05	05	05	010	05	05	07	05	05	05	05	05	No
D5	{1}	05	05	05	05	05	05	05	05	05	05	05	05	05	No
D6	{12, 13, 14, 15, 17, 19, 21, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D7	{12, 13, 15, 16, 17, 19, 21, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D8	{12, 13, 15, 17, 18, 19, 21, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No
D9	{12, 13, 15, 17, 19, 20, 21, 22}	06	07	08	09	010	05	05	05	05	05	05	05	05	No